

ViLL-E: Video LLM Embeddings for Retrieval

ViLL-E: 用于检索的视频 LLM 嵌入向量

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Abstract

摘要

Video Large Language Models (VideoLLMs) excel at video understanding tasks where outputs are textual, such as Video Question Answering and Video Captioning. However, they underperform specialized embedding-based models in Retrieval tasks, such as Text-to-Video Retrieval and Moment Retrieval. We introduce ViLL-E (Video-LLM-Embed), a unified VideoLLM architecture endowed with a novel embedding generation mechanism that allows the model to “think longer” for complex videos and stop early for easy ones. We train this model with a three-stage training methodology combining generative and contrastive learning: initial large-scale pre-training with video-caption pairs; followed by continual training on a smaller, detailed-caption dataset; and concluding with task-specific fine-tuning on a novel multi-task dataset covering Video QA, Temporal Localization, Video Retrieval, and Video-Text Matching. Our model significantly improves temporal localization (on avg. 7% over other VideoLLMs) and video retrieval (up to 4% over dual encoder models), achieving performance comparable to state-of-the-art specialized embedding models while remaining competitive on VideoQA tasks. Furthermore, our joint contrastive-generative training unlocks new zero-shot capabilities, significantly outperforming state-of-the-art methods in composed video retrieval (+5% over SotA) and retrieval from long text (+2% over SotA).

视频大语言模型（VideoLLMs）在输出为文本的视频理解任务中表现出色，例如视频问答和视频描述生成。然而，在检索任务中，如文本到视频检索和时刻检索，它们的表现不如专门的嵌入式模型。我们引入了 ViLL-E（Video-LLM-Embed），这是一种统一的 VideoLLM 架构，配备了新颖的嵌入生成机制，使模型能够对复杂视频进行“更长时间的思考”，并对简单视频提前停止处理。我们采用一种结合生成和对比学习的三阶段训练方法来训练该模型：首先进行大规模的视频-描述对预训练；接着在较小的详细描述数据集

上进行持续训练；最后在涵盖视频问答、时间定位、视频检索和视频-文本匹配的新型多任务数据集上进行任务特定的微调。我们的模型在时间定位方面显著提升（平均超过其他 VideoLLMs 7%），并在视频检索方面也有显著提升（比双编码器模型高出最多 4%），在保持视频问答任务竞争力的同时，其性能与最先进的专用嵌入模型相当。此外，我们的联合对比-生成训练解锁了新的零样本能力，在组合视频检索（比当前最佳方法高 5%）和从长文本中检索（比当前最佳方法高 2%）方面显著优于现有方法。

1 Introduction

1 引言

Large Language Models (LLMs) have demonstrated significant capabilities in a “single model multi-task approach” setting, effectively solving various natural language understanding tasks. Inspired by the success of LLMs [7, 41, 13, 34], numerous studies have integrated additional modalities like images [19, 26, 3] and audio [38, 56] into these models. These Multimodal LLMs models excel at most vision tasks that can be formulated as text generation problems. Of specific interest are Video Large Language Models (VideoLLMs) [29, 24, 57, 21], which incorporate videos into LLMs by encoding video frames and aligning visual features with LLM feature spaces via projection layers, enabling video understanding through tasks like captioning and QA.

大型语言模型（LLMs）在“单模型多任务”设置中展现出显著的能力，能够有效解决各种自然语言理解任务。受 LLMs 成功启发 [7, 41, 13, 34]，大量研究将额外的模态如图像 [19, 26, 3] 和音频 [38, 56] 整合到这些模型中。这些多模态 LLM 模型在大多数可以表述为文本生成问题的视觉任务上表现出色。特别值得关注的是视频大语言模型（VideoLLMs）[29, 24, 57, 21]，它们通过编码视频帧并将视觉特征与 LLM 特征空间对齐（通过投影层），将视频整合到 LLMs 中，从而通过字幕生成和问答等任务实现视频理解。

However, this generative approach has limitations. Many video tasks, like Text-to-Video Retrieval (T2V), require embedding-based matching rather than text generation. Such tasks typically use specialized models with aligned video-text embeddings, fine-tuned per dataset. Currently these specialized methods lead performance on standard benchmarks. Even VideoLLM variants that are specialized to attempt temporal reasoning (e.g. LLaVA-ST [18]) trail expert detectors such as QD-DETR [30] by double-digit margins on ActivityNet [15] and Charades-STA [10]. Figure 1 underscores the point: every top performer in video retrieval, temporal localisation or cross-modal search (VidLA [37], QD-DETR [30], SigLIP [53]) achieves its performance with embeddings, while Video-LLMs dominate the Video-QA column. Also, recent NLP research has demonstrated that LLMs can be transformed into strong retrieval models with contrastive finetuning on surprisingly little data (see, e.g., GRIT [31] or E5 [43]). This raises the question: Instead of maintaining two separate stacks, can a single model handle both generative tasks and extract discriminative video/text embeddings, addressing the retrieval and localization gaps?

然而，这种生成方法存在一定的局限性。许多视频任务，例如文本到视频检索（T2V），需要基于嵌入的匹配，而不是文本生成。这类任务通常使用专门的模型，这些模型具有对齐的视频-文本嵌入，并针对特定数据集进行微调。目前，这些专门的方法在标准基准测试中表现优异。即使是专门用于尝试时间推理的 VideoLLM 变体（例如 LLaVA-ST [18]），在 ActivityNet [15] 和 Charades-STA [10] 等数据集上，也比专家检测器（如 QD-DETR [30]）落后数十个百分点。图 1 说明了这一点：在视频检索、时间定位或跨模态搜索（如 VidLA [37]、QD-DETR [30]、SigLIP [53]）中表现最佳的模型，都是通过嵌入实现其性能的，而 Video-LLMs 在视频问答（Video-QA）列中占据主导地位。此外，最近的自然语言处理（NLP）研究表明，通过在极少量数据上进行对比微调，可以将大型语言模型（LLMs）转变为强大的检索模型（例如 GRIT [31] 或 E5 [43]）。这引发了一个问题：与其维护两个独立的模型栈，是否可以使用一个模型同时处理生成任务和提取具有判别性的视频/文本嵌入，从而解决检索和定位方面的差距？

Motivated by these observations, we propose an approach that combines generative and embedding tasks during Video-LLM training, aiming for strong performance on generative tasks like VideoQA, while also enabling embedding generation for tasks such as video retrieval—previously unsupported by VideoLLMs. Our approach builds on PaliGemma [6], a standard multi-modal LLM

受这些观察的启发，我们提出了一种在视频-语言模型（Video-LLM）训练过程中结合生成任务和嵌入任

务的方法，旨在在像视频问答（VideoQA）这样的生成任务中实现强大的性能，同时也能支持诸如视频检索等任务中的嵌入生成——这些功能此前在 VideoLLMs 中并未得到支持。我们的方法基于 PaliGemma [6]，这是一个标准的多模态语言模型

VidLA (Video-Text Retrieval Expert)	Qwen2.5-VL-3B (General Video LLM)
SigLIP (Image-Text Retrieval Expert)	LLaVA-ST-7B (Temporal Video LLM)
QD-DETR (Localization Expert)	Ours

VidLA (视频-文本检索专家)	Qwen2.5-VL-3B (通用视频大模型)
SigLIP (图像-文本检索专家)	LLaVA-ST-7B (时间序列视频 LLM)
QD-DETR (定位专家)	我们的

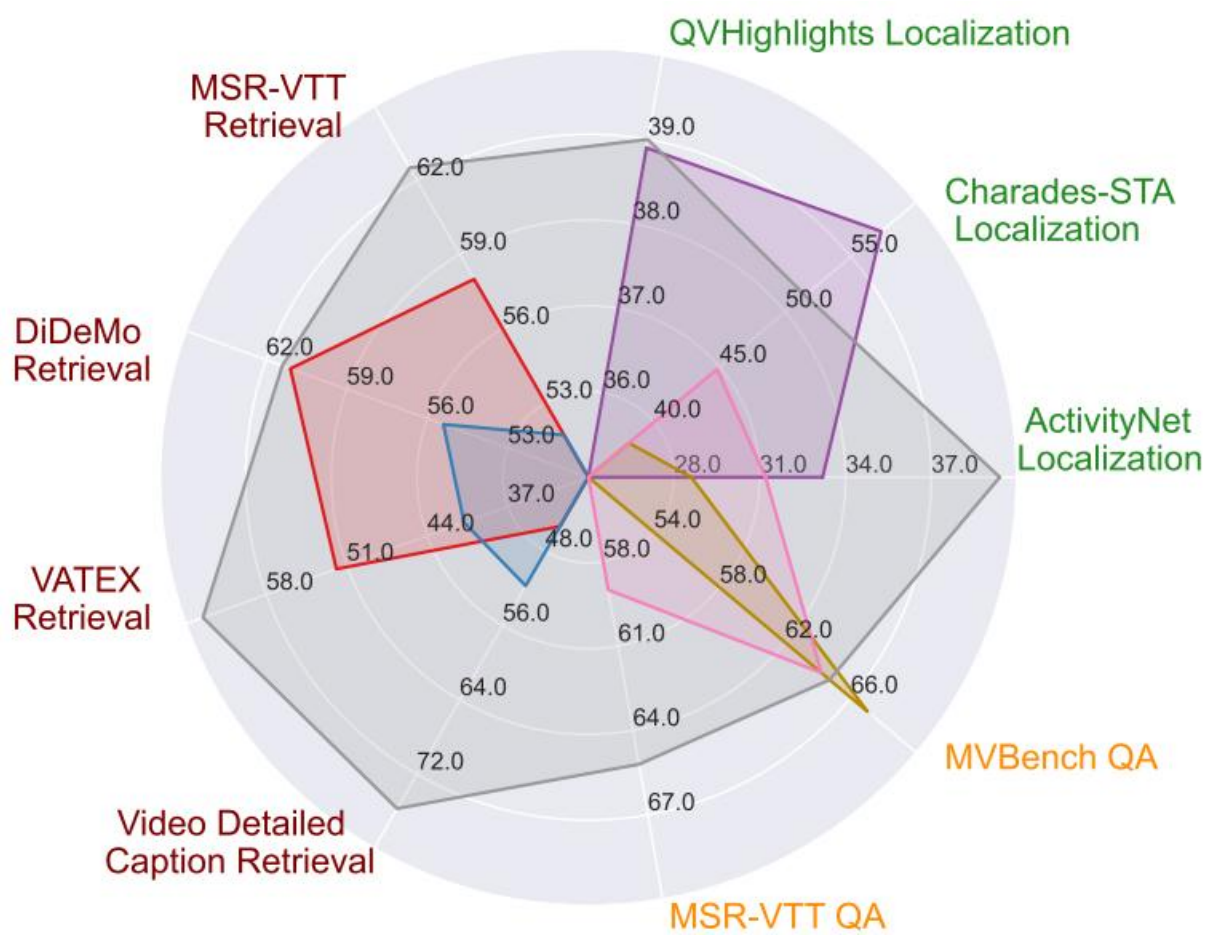


Figure 1: images/9745263ee4e22508caf52dfe7f78fc20c312aa0cf2507654ab34c1cb350a1d39.jpg

Method	Architecture	
LLM	Embeddings	
VidLA	□	□
SigLIP	□	□

QD-DETR	□	□
Qwen2.5-VL	□	□
LLaVA-ST	□	□
Ours	□	□

方法	架构	
大型语言模型	嵌入向量	
VidLA	□	□
SigLIP	□	□
QD-DETR	□	□
Qwen2.5-VL	□	□
LLaVA-ST	□	□
我们的	□	□

Method	Capabilities			
Ret.	Loc.	QA	Cap.	
VidLA	□	□	□	□
SigLIP	□	□	□	□
QD-DETR	□	□	□	□
Qwen2.5-VL	□	□	□	□
LLaVA-ST	□	□	□	□
Ours	□	□	□	□

方法	能力			
退休	位置	问答	章	
VidLA	□	□	□	□
SigLIP	□	□	□	□
QD-DETR	□	□	□	□
Qwen2.5-VL	□	□	□	□
LLaVA-ST	□	□	□	□
我们的	□	□	□	□

Figure 1: Left & Bottom Right: VideoLLMs lag expert models on some retrieval-based tasks, e.g. Temporal Localization (in green [17, 10, 15]), and are incapable of others, e.g. Text-to-Video Retrieval (in red [49, 1, 44]). A key difference between existing VideoLLMs and state of the art expert models in these tasks is the use of embeddings(Top Right). Our approach, ViLL-E (VideoLLM-Embed, pronounced willy) equips VideoLLMs with embedding generation. ViLL-E matches SotA on retrieval tasks while remaining competitive on generation tasks such as VideoQA (in orange [47, 22]). □ indicates limited capabilities.

图 1: 左 & 底右: 在一些基于检索的任务中, VideoLLMs 的表现落后于专家模型, 例如时间定位 (以绿色 [17, 10, 15] 表示), 并且无法完成其他任务, 例如文本到视频检索 (以红色 [49, 1, 44] 表示)。现有

VideoLLMs 与这些任务中的最先进专家模型之间的一个关键区别在于使用了嵌入（右上）。我们的方法 ViLL-E (VideoLLM-Embed, 发音为 willy) 为 VideoLLMs 配备了嵌入生成能力。ViLL-E 在检索任务上达到了最先进的水平，同时在生成任务（如视频问答，以橙色 [47, 22] 表示）上仍保持竞争力。□ 表示能力有限。

and adds a learnable attentive-pooling “embedding head” after the LLM decoder. The embedding head is designed to adapt to video complexity during inference. This allows the model to “think longer” for hard videos but return quickly for easy ones. We propose three-stage training, which begins with large-scale pre-training on video-caption pairs using joint supervision: next-word prediction for generation and a multi-modal contrastive loss for embedding. In the second stage, we continue pre-training on a smaller dataset of videos with high-quality generated captions. The final stage involves instruction finetuning on a multi-task dataset covering VideoQA, captioning, localization, and retrieval. Our proposed model, ViLL-E (pronounced willy), is competitive with VideoLLMs on VideoQA and outperforms them on localization. Our embeddings narrow the gap with non-LLM localization expert methods and enable SoTA performance on video retrieval.

并在 LLM 解码器后添加了一个可学习的注意力池化“嵌入头” (embedding head)。该嵌入头设计用于在推理过程中适应视频的复杂性。这使得模型在处理困难视频时能够“思考更久”，而在处理简单视频时能够快速返回。我们提出了三阶段的训练方法，第一阶段是使用联合监督在视频-描述对上进行大规模预训练：用于生成的下一个词预测，以及用于嵌入的多模态对比损失。第二阶段则继续在具有高质量生成描述的小型视频数据集上进行预训练。第三阶段则是在涵盖视频问答 (VideoQA)、描述生成、定位和检索的多任务数据集上进行指令微调。我们提出的模型 ViLL-E（发音为“willy”）在视频问答任务上与 VideoLLMs 相当，并在定位任务上表现优于它们。我们的嵌入方法缩小了与非 LLM 定位专家方法之间的差距，并在视频检索任务上实现了最先进的性能。

In summary, the key contributions of our work are:

总之，我们的工作主要贡献如下：

- We present ViLL-E, an unified solution to diverse video-understanding tasks, which is the first VideoLLM that can generate text responses or video & text embeddings with a single model.
- An effective three-stage training pipeline: (1) large-scale joint captioning + contrastive learning, (2) continued pre-training on high quality captions, and (3) multitask fine-tuning.
- 我们提出了 ViLL-E，这是一个统一的解决方案，适用于各种视频理解任务，它是首个能够通过单一模型生成文本回复或视频与文本嵌入的 VideoLLM。
- 一个有效的三阶段训练流程：(1) 大规模联合字幕生成 + 对比学习

ing, (2) continued pre-training on high quality captions, and (3) multitask fine-tuning.

ing, (2) 在高质量字幕上继续预训练，以及 (3) 多任务微调。

- Our single model ViLL-E, outperforms VideoLLMs by more than 7% across three moment retrieval benchmarks, rivals them on VideoQA, and is competitive with expert retrieval models.
- ViLL-E also solves new tasks in zero-shot setting, including long-caption retrieval, compositional video search, and two-stage retrieve-and-match pipelines using a single model.
- 我们的单模型 ViLL-E 在三个时刻检索基准测试中表现优于 VideoLLMs 超过 7%，在视频问答 (VideoQA) 任务中与之竞争，并且与专家检索模型具有竞争力。
- ViLL-E 还在零样本设置下解决了新任务，包括长字幕检索、组合视频搜索，以及使用单一模型的两

阶段检索-匹配流程。

2 Related Work

2 相关工作

Video Expert Models for Retrieval and Localization: Prior works on text-video and video-text retrieval [50, 28, 11, 37, 45] typically extend dual-encoder image-text models such as CLIP [33] or BLIP [20] to video by aggregating the CLIP vision encoder outputs from multiple frames with architecture adaptations to model temporal relations [5, 8]. For retrieval they use dense vector search in the embedding space while a few other methods use an additional reranking step with cross-modality fusion layers [23, 9]. Prior works on temporal localization such as Moment-DETR [17], QD-DETR [30], TR-DETR [39] and UnLoc [51] build on ideas from retrieval. Typically Video-Text Fusion is also employed to score the match between a given text and video segment for compatibility. Specific unique

视频检索与定位的专家模型：先前关于文本-视频和视频-文本检索的工作 [50, 28, 11, 37, 45] 通常通过在架构上进行调整以建模时间关系 [5, 8]，将双编码器图像-文本模型如 CLIP [33] 或 BLIP [20] 扩展到视频，方法是通过聚合多帧 CLIP 视觉编码器的输出。对于检索，它们在嵌入空间中使用密集向量搜索，而一些其他方法则使用额外的重排序步骤，并结合跨模态融合层 [23, 9]。先前关于时间定位的工作，如 Moment-DETR [17]、QD-DETR [30]、TR-DETR [39] 和 UnLoc [51]，则借鉴了检索的思路。通常也采用视频-文本融合来评估给定文本与视频片段之间的匹配度以判断其兼容性。具体独特的

ideas include additional modules like time span prediction in QD-DETR and learnable moment queries in Moment-DETR. Vid2Seq [52] utilizes dense captioning trained on large video-text datasets instead of retrieval for localization. These expert models are close to the state of the art frontier for their respective tasks, however they are specialized for one task, and are typically finetuned individually for each dataset.

想法包括额外的模块，例如 QD-DETR 中的时间跨度预测和 Moment-DETR 中的学习时刻查询。Vid2Seq [52] 利用在大型视频-文本数据集上训练的密集字幕技术，而不是使用检索来进行定位。这些专家模型在各自的任務上接近当前的前沿水平，然而它们专门针对某一任务，并且通常为每个数据集单独进行微调。

Video LLMs: Many recent works have extended ImageLLMs such as LLaVA [26] to videos. VideoLLaVA [24] is a straightforward extension that utilizes LanguageBind [59] encoder instead of CLIP because its trained on both images and videos. Some works have tried to extend a BLIP2 style architecture to Videos, namely VideoChat [21] and Video LLaMA [57]. In addition, some works have focused on creation of strong IFT datasets specific to video tasks, VideoChatGPT [29] created IFT data using ActivityNet dense captions and semi-automated process with questions generated by GPT3.5 using dense text captions and open vocabulary attributes. In VideoChat2 [21], another diverse IFT dataset was created by combining many (30+) datasets together. These models excel at video question answering and other text generation tasks, however they often struggle at temporal localization tasks and don't generate embeddings, which are necessary for video retrieval.

视频大模型：许多最近的工作将图像大模型如 LLaVA [26] 扩展到了视频领域。VideoLLaVA [24] 是一个直接的扩展，它使用 LanguageBind [59] 编码器而不是 CLIP，因为其在图像和视频上都进行了训练。一些工作尝试将 BLIP2 风格的架构扩展到视频，即 VideoChat [21] 和 Video LLaMA [57]。此外，一些工作专注于创建特定于视频任务的强大 IFT 数据集，例如 VideoChatGPT [29] 利用 ActivityNet 密集字幕和通过 GPT3.5 生成的问题进行半自动化处理来创建 IFT 数据。在 VideoChat2 [21] 中，通过结合多个 (30 多个) 数据集创建了另一个多样化的 IFT 数据集。这些模型在视频问答和其他文本生成任务上表现出色，但它们通常在时间定位任务上表现不佳，并且无法生成嵌入，这对于视频检索是必要的。

Video LLMs with Localization IFT: Recent works have sought to instill temporal awareness and localization skills in video LLMs. Broadly, the architectures of these approaches can be categorized into LLaVA-like or BLIP2-like [19] depending on whether or not they use a Q-Former for token reduction before input to LLM. TimeChat [36] uses sliding window Q-Formers as a projection layer,

VTimeLLM [12] uses MLP, HawkEye [46] uses simple Q-Former and Momentor [32] proposes a parallel temporal perception module as part of its projection layers. LLaVA-ST [18] inserts special spatio-temporal tokens into the language stream, and aligns them with vision features through a language-aligned position embedding. To impart temporal awareness to the models, these works use IFT datasets with specialized tasks such as dense captioning, timestamp generation for localization, repetition counting and more. On localization tasks, these models outperform general VideoLLMs, however they are unable to outperform expert models which do not use LLMs.

视频 LLMs 与定位 IFT: 最近的研究尝试在视频 LLMs 中引入时间感知和定位能力。总体而言, 这些方法的架构可以根据是否在输入 LLM 之前使用 Q-Former 进行 token 缩减, 分为 LLaVA-like 或 BLIP2-like [19]。TimeChat [36] 使用滑动窗口 Q-Formers 作为投影层, VTimeLLM [12] 使用 MLP, HawkEye [46] 使用简单的 Q-Former, 而 Momentor [32] 在其投影层中提出了一种并行时间感知模块。LLaVA-ST [18] 将特殊的时空 token 插入语言流中, 并通过语言对齐的位置嵌入将它们与视觉特征对齐。为了向模型传授时间感知, 这些工作使用了包含专门任务 (如密集描述、定位的时间戳生成、重复计数等) 的 IFT 数据集。在定位任务上, 这些模型优于一般的 VideoLLMs, 但它们无法超越不使用 LLMs 的专家模型。

LLMs for Embedding generation: Some works in the NLP domain [43, 16] have demonstrated that generative LLMs can be finetuned with a limited amount of data (relative to the scale of generative pre-training) to generate strong embeddings for a variety of downstream tasks including retrieval. GRIT [31] has demonstrated that a pretrained LLM can be jointly finetuned to improve on both generative instruction following and embedding generation, maintaining strong performance on both sets of tasks. Some recent concurrent works such as VLM2Vec [14], GME [58] and MM-Embed [25] have sought to use LLMs for multimodal embeddings, but are limited to images.

用于嵌入生成的 LLMs: 一些在 NLP 领域的工作 [43, 16] 表明, 生成式 LLMs 可以使用相对较少的数据 (相对于生成式预训练的规模) 进行微调, 以生成适用于多种下游任务 (包括检索) 的强大嵌入。GRIT [31] 证明了预训练的 LLM 可以联合微调, 从而在生成式指令遵循和嵌入生成两个方面都得到提升, 并在两项任务上保持良好的性能。一些最近的并行工作, 如 VLM2Vec [14]、GME [58] 和 MM-Embed [25], 试图使用 LLMs 生成多模态嵌入, 但目前仅限于图像。

Our approach unifies text-generative and embedding-generative capabilities into a single VideoLLM model, which is competitive with task-specific and dataset-specific expert models on temporal localization and retrieval tasks that require embedding generation, as well as with Video LLMs on text generation like QA and captioning tasks.

我们的方法将文本生成和嵌入生成能力统一到一个 VideoLLM 模型中, 该模型在需要嵌入生成的时序定位和检索任务中, 与专门针对特定任务和数据集的专家模型具有竞争力, 同时在文本生成任务 (如问答和字幕生成) 中, 也与 Video LLMs 相当。

3 Method

3 方法

In this section, we present the architecture and training methodology of our VideoLLM, designed to leverage multi-modal video data to perform both generative and embedding tasks. Our approach involves a multi-stage and multi-task process that includes continued multi-modal pre-training to improve video language alignment, and subsequent multi-task finetuning.

在本节中, 我们介绍了 VideoLLM 的架构和训练方法, 该模型旨在利用多模态视频数据执行生成任务和嵌入任务。我们的方法包括一个分阶段、多任务的处理过程, 包括持续的多模态预训练以提高视频语言对齐, 以及随后的多任务微调。

3.1 Model Architecture

3.1 模型架构

Our model (see Figure 2) features an LLM backbone, shared by both its generative and embedding tasks, integrated with a vision encoder module designed to effectively process video inputs. We initialize the model's weights using a pretrained multimodal LLM (PaliGemma-3B [6]). Following PrefixLM (Raffel et al., 2020) [34], visual tokens and input prompts are processed with bidirectional attention, whereas causal attention is applied to the suffix autoregressively generated by the model. For text generation tasks (e.g. VideoQA), our model follows the same approach as PaliGemma.

我们的模型（参见图 2）采用了一个大语言模型（LLM）主干，该主干同时用于生成任务和嵌入任务，并与一个专门设计用于有效处理视频输入的视觉编码模块集成。我们使用一个预训练的多模态 LLM (PaliGemma-3B [6]) 来初始化模型的权重。遵循 PrefixLM (Raffel 等, 2020) [34], 视觉标记和输入提示通过双向注意力进行处理, 而因果注意力则应用于模型自回归生成的后缀。对于文本生成任务（例如 VideoQA），我们的模型采用了与 PaliGemma 相同的处理方式。

3.2 Embedding Head Design

3.2 嵌入头设计

We introduce a learnable embedding head to support retrieval-oriented tasks with four design goals: (1) dedicated modeling capacity separate from generative tasks, (2) a bottleneck that yields dense embeddings without overly restricting representation power, (3) adaptability to variable token lengths for videos of different complexity, and

我们引入一个可学习的嵌入头部，以支持面向检索的任务，其设计目标包括以下四个：(1) 与生成任务分离的专用建模能力，(2) 一个瓶颈结构，能够在不过度限制表示能力的情况下生成密集嵌入，(3) 对不同复杂度视频的可变标记长度的适应性，以及

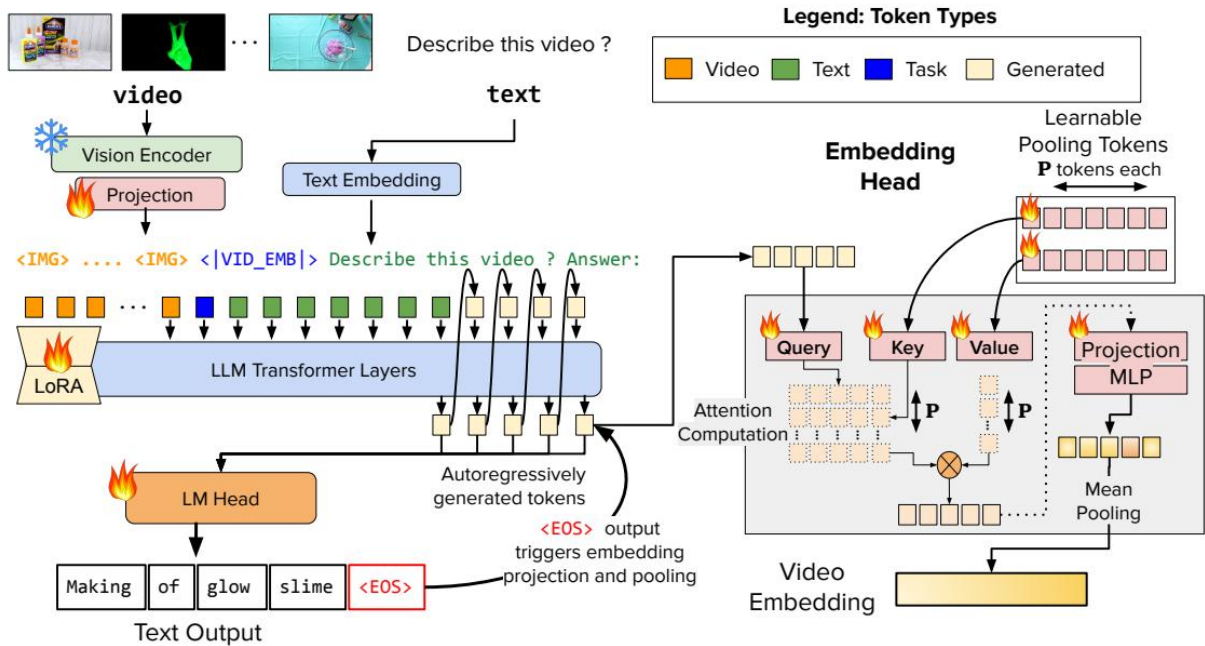


Figure 2: images/1568c6286b02224dcc5d71afbf689fb37667c07c2a2d20b2fc98b109b367e8c7.jpg

Figure 2: Our model ViLL-E is a multi-modal LLM which has been equipped with additional embedding generation capabilities. Individual video frames are first encoded by a pretrained vision encoder, with extracted features projected into embedding vectors. These visual embeddings are combined with textual embeddings of an input prompt and jointly processed autoregressively through a pre-trained Multi-Modal LLM’s transformer layers. After the model autoregressively generates an end-of-sequence ($|\text{EOS}|$) token, all the generated tokens are collected and passed to our embedding head. The embedding head employs an attention-based learnable pooling mechanism to extract and aggregate relevant information across the tokens; The generated tokens are further passed through projection layers and mean pooled to yield a compact, informative video embedding. The ability to dynamically generate different number of tokens to feed the embedding head allows the model to handle videos of different complexity levels and is key for its success at varied retrieval tasks.

图 2: 我们的模型 ViLL-E 是一个多模态大语言模型, 其具备额外的嵌入生成能力。首先, 单个视频帧通过一个预训练的视觉编码器进行编码, 提取的特征被投影到嵌入向量中。这些视觉嵌入与输入提示的文本嵌入相结合, 并通过预训练的多模态大语言模型的 Transformer 层进行联合自回归处理。在模型自回归地生成一个序列结束标记 ($|\text{EOS}|$) 后, 所有生成的标记都会被收集并传递到我们的嵌入头。嵌入头采用基于注意力的可学习池化机制, 以跨标记提取和聚合相关信息; 生成的标记进一步通过投影层和均值池化处理, 以生成一个紧凑且信息丰富的视频嵌入。模型能够动态生成不同数量的标记以喂入嵌入头, 使其能够处理不同复杂程度的视频, 并在各种检索任务中取得成功。

(4) parameter efficiency. We evaluated several approaches: 1. Attention-Free pooling: Pools penultimate layer LLM tokens into a fixed-size embedding, lightweight but dependent on heuristic pooling (cf. Sentence-BERT [35]). 2. Self-Attention: Adds an extra transformer layer, offering some task-specific capacity but redundant with prior LLM layers and lacking a bottleneck. 3. Q-Former: Learned queries cross-attend to LLM outputs to produce fixed-length embeddings (e.g., BLIP-2 [19]); effective but inflexible for variable content length, as output size always equals number of learnable queries. 4. ‘K-Former’: Learns only keys, enabling variable-length outputs. 5. Our ‘KV-Former’: Uses LLM tokens as queries with P learnable key and value “pooling tokens,” each. Enabling variable-length outputs while acting as a bottleneck akin to dictionary learning.

(4) 参数效率。我们评估了多种方法: 1. 无注意力池化: 将 LLM 的倒数第二层 token 池化为固定大小的嵌入, 轻量但依赖启发式池化 (参见 Sentence-BERT [35])。2. 自注意力: 添加一个额外的 transformer 层, 提供一些任务特定的能力, 但与之前的 LLM 层冗余, 并且缺乏瓶颈。3. Q-Former: 学习的查询通过交叉注意力机制关注 LLM 的输出, 以生成固定长度的嵌入 (例如, BLIP-2 [19]); 有效但对内容长度变化不够灵活, 因为输出大小始终等于可学习查询的数量。4. ‘K-Former’: 仅学习键, 从而实现可变长度的输出。5. 我们的‘KV-Former’: 使用 LLM 的 token 作为查询, 每个都带有 P 个可学习的键和值“池化 token”。这使得输出具有可变长度, 同时起到类似于字典学习的瓶颈作用。

Our embedding head (on the right in Fig. 2) receives tokens from the final LLM transformer layer and employs a learnable attentive pooling mechanism. Specifically, a transformer-based self-attention layer adaptively weighs tokens, using the LLM’s tokens as queries and P learnable keys and values (“pooling tokens”). After this attention operation, the weighted representations are projected

我们的嵌入头 (见图 2 右侧) 接收来自最终 LLM transformer 层的 token, 并采用一种可学习的注意力池化机制。具体来说, 一个基于 transformer 的自注意力层会自适应地对 token 进行加权, 使用 LLM 的 token 作为查询, P 个可学习的键和值 (“池化 token”)。在完成这一注意力操作后, 加权表示会被投影

into the embedding space through a multi-layer perceptron (MLP) and subsequently mean-pooled.

通过一个多层感知机 (MLP) 进入嵌入空间, 随后进行均值池化。

For the video embedding generation, our model uniquely leverages an end-of-sequence (O)-triggered embedding generation mechanism. This mechanism allows the model to utilize variable inference steps tailored to each video’s complexity - more steps for complex videos needing deeper analysis, fewer for simple ones. This adaptive computation approach substantially improves performance over existing fixed-step LLM-based embedding methods, offering

both efficiency and improved representational quality. The proposed design is flexible and can be leveraged for various tasks, e.g. text and video embeddings for temporal localization, composed video retrieval etc. For text embedding generation, we rely on standard inference similar to prior works [31], with entire text embeddings being generated in a single step.

对于视频嵌入生成，我们的模型独特地利用了由序列结束标记 $\langle \rangle$ 触发的嵌入生成机制。这种机制使得模型能够根据每个视频的复杂度使用可变的推理步骤——复杂视频需要更多的步骤进行深入分析，而简单视频则使用较少的步骤。这种自适应计算方法在性能上显著优于现有的基于固定步长的 LLM 嵌入方法，既提高了效率，又改善了表示质量。所提出的这种设计具有灵活性，可以用于各种任务，例如用于时间定位的文本和视频嵌入，以及组合视频检索等。对于文本嵌入生成，我们依赖于类似于之前工作的标准推理方法 [31]，整个文本嵌入可以在一步中生成。

3.3 Training Objectives

3.3 训练目标

We follow a three-stage training strategy: (1) pre-training on a large, diverse, and noisy dataset to jointly improve video captioning and embedding capabilities; (2) continued pre-training on a smaller, higher-quality dataset; and (3) multi-task finetuning to unlock the model’s performance across diverse

我们采用一种三阶段的训练策略：(1) 在一个大规模、多样且包含噪声的数据集上进行预训练，以共同提升视频描述和嵌入能力；(2) 在一个规模较小但质量更高的数据集上继续预训练；以及 (3) 多任务微调，以在多样化的场景中释放模型的性能

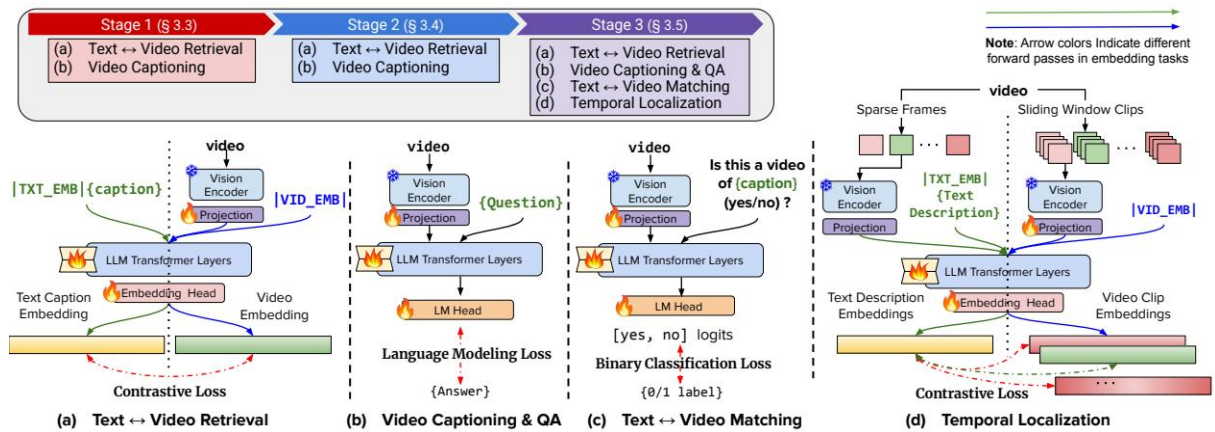


Figure 3: images/85c1f6591e8f39375cc74d7eeadd0c4dc0e5f4238aade4dbd093e92e3179d81e.jpg

Figure 3: Three stages of training. Stage 1 (§ 3.4), the large scale joint contrastive and generative pre-training, which utilizes (a) Video-Text retrieval task and (b) Video captioning on the large scale Shutterstock video dataset. Stage 2 (§ 3.5), continues a similar joint pre-training approach (retrieval + captioning tasks as shown in (a) and (b)) on a smaller high quality dataset created using Claude-3-Sonnet. Stage 3 (§ 3.6), our multi-task fine-tuning stage, in addition to the prior tasks (retrieval + captioning) integrates (b) Video QA, (c) Video-Text matching and (d) Temporal Localization to improve all-around retrieval capabilities.

图 3：训练的三个阶段。阶段 1 (§ 3.4)，大规模联合对比学习和生成式预训练，利用 (a) 视频-文本检索任务和 (b) 在大规模 Shutterstock 视频数据集上进行的视频字幕生成。阶段 2 (§ 3.5)，继续使用类似的联合预训练方法（检索 + 字幕生成任务，如 (a) 和 (b) 所示）在一个使用 Claude-3-Sonnet 创建的小规模高质量数据集上进行训练。阶段 3 (§ 3.6)，我们的多任务微调阶段，除了之前的任务（检索 + 字幕生成）外，还整合了 (b) 视频问答、(c) 视频-文本匹配和 (d) 时间定位，以提高全面的检索能力。

downstream tasks. To ensure parameter efficiency during fine-tuning, we adopt Low-Rank Adapters (LoRA). The vision projection module and embedding head are fully trained to maximize representation learning. Our model is designed to effectively balance multimodal generative capabilities with state-of-the-art performance on embedding-oriented retrieval tasks. Across our training stages, a total of four different tasks are used, which are described below.

下游任务。为了在微调过程中确保参数效率，我们采用低秩适配器（LoRA）。视觉投影模块和嵌入头完全训练以最大化表征学习。我们的模型旨在有效平衡多模态生成能力与面向嵌入的检索任务上的最先进性。在我们的训练阶段中，总共使用了四种不同的任务，如下所述。

Text-Video Retrieval (Fig. 3a). This task aligns video and text embeddings using CLIP-like [33] contrastive loss, encouraging each paired example to outrank in-batch negatives.

文本-视频检索（图 3a）。该任务使用类似 CLIP 的对比损失 [33] 对视频和文本嵌入进行对齐，鼓励每对示例在批次内的负样本中排名更高。

Video Captioning & Question-Answering (Fig. 3b). The goal of captioning is for the model to generate accurate descriptions of videos. The model learns to predict each token in the caption, given the previous tokens and the video. For QA, the model generates answers to questions about the video; the question serves as part of the prompt, guiding the model’s attention to answer accurately. We use the next-token prediction loss.

视频字幕生成与问答（图 3b）。字幕生成的目标是让模型生成准确的视频描述。模型学习根据之前的标记和视频内容来预测字幕中的每个标记。对于问答任务，模型生成关于视频的问题的答案；问题作为提示的一部分，引导模型的注意力以准确回答。我们使用下一个标记预测损失。

Text-Video Matching (Fig. 3c). As a complement to the retrieval task we also formulate a matching task where the model is trained to identify if a given video and caption form a matching pair. Binary decisions are supervised by labels.

文本-视频匹配（图 3c）。作为检索任务的补充，我们还提出了一种匹配任务，其中模型被训练以确定给定的视频和描述是否构成匹配对。二元决策由标签进行监督。

Temporal Localization (Fig. 3d). Localization requires the model to match specific video segments to text annotations while distinguishing similar but incorrect segments. Sliding window hard negative mining provides challenging non-matching segments with limited temporal overlap ($\text{IoU} < 0.2$),

时间定位（图 3d）。定位要求模型将特定的视频片段与文本注释进行匹配，同时区分相似但错误的片段。滑动窗口硬负样本挖掘提供了具有有限时间重叠（ $\text{IoU} < 0.2$ ）的具有挑战性的不匹配片段，

helping the model learn to differentiate similar content. We use a standard contrastive formulation with one positive and two hard negatives per query. Complete notation and full equations are provided in the Appendix B. The following subsections discuss our multi-stage training strategy in detail.

帮助模型学习区分相似内容。我们使用标准的对比学习公式，每个查询包含一个正样本和两个困难负样本。完整的符号和完整公式在附录 B 中提供。以下小节详细讨论了我们的多阶段训练策略。

3.4 Training Stage 1: Generative-Contrastive Pre-Training

3.4 训练阶段 1：生成-对比预训练

In stage 1, we utilize a large video-caption dataset and jointly train the network using generative and contrastive supervision. In the first forward pass, for each video caption pair, video input tokens are passed along with a prompt asking the model to describe the video. Captioning output for the generative part and the video embedding are generated simultaneously in this

pass. This is followed by a second computationally lighter forward pass with text-only input to generate the text embedding. The next token prediction loss is applied to the caption generation task, and the CLIP multimodal contrastive loss is applied for the embedding generation task. The generative component aims to improve the model's ability to create coherent and contextually relevant textual descriptions of the video content, while the contrastive component enhances its discriminative power by teaching it to differentiate between relevant and irrelevant text-video pairs. This dual-objective training strategy ensures that the model not only learns to generate high-quality textual content but also develops a better alignment between modalities.

在第一阶段，我们利用一个大型的视频-字幕数据集，并通过生成式和对比监督联合训练网络。在第一次前向传递中，对于每个视频-字幕对，视频输入标记会与一个提示一起传递给模型，要求模型描述视频。在此过程中，生成式部分的字幕生成输出和视频嵌入会同时生成。随后进行一次计算量更小的仅文本输入的前向传递，以生成文本嵌入。接下来的下一个标记预测损失应用于字幕生成任务，而 CLIP 多模态对比损失则应用于嵌入生成任务。生成式部分旨在提高模型创建连贯且与上下文相关的视频内容文本描述的能力，而对比部分则通过教会模型区分相关和不相关的文本-视频对，从而增强其判别能力。这种双目标训练策略确保模型不仅学会生成高质量的文本内容，还能更好地实现模态之间的对齐。

3.5 Training Stage 2: Continued Pre-Training

3.5 训练阶段 2：持续预训练

While our pre-training data has a diverse collection of videos, the captions tend to be short summaries

虽然我们的预训练数据包含多样化的视频，但字幕往往只是简短的总结

of the video content, often missing many details in the video. Since preserving important details from the video in our VideoLLM's feature space is beneficial for downstream tasks, we supplement our pre-training with a small intermediate pre-training phase using higher quality captions for a subset of videos from our pre-training set. The dataset for intermediate pre-training with high quality descriptive captions is discussed in detail in Section 4.

关于视频内容，经常遗漏许多视频细节。由于在 VideoLLM 的特征空间中保留视频中的重要细节对下游任务是有益的，我们使用预训练集一部分视频的高质量字幕进行一个小型的中间预训练阶段，以补充我们的预训练。关于使用高质量描述性字幕进行中间预训练的数据集的详细讨论，请参见第 4 节。

3.6 Training Stage 3: Multi-Task Finetuning

3.6 训练阶段 3：多任务微调

In this stage, a high-quality finetuning dataset is used combining supervision across four tasks: captioning or question answering (QA), retrieval, matching and temporal localization. The finetuning process integrates these tasks to ensure that the model develops well-rounded capabilities, enabling it to handle diverse real-world scenarios that involve multi-modal understanding and generation.

在这个阶段，使用了一个高质量的微调数据集，结合了四个任务的监督：描述生成或问答（QA）、检索、匹配和时间定位。微调过程将这些任务整合在一起，确保模型能够发展出全面的能力，从而使其能够处理涉及多模态理解和生成的各种现实世界场景。

4 Training Data

4 训练数据

In this section, we provide details about the datasets used in the different training stages of our model.

在本节中，我们提供了有关我们模型不同训练阶段所使用的数据集的详细信息。

4.1 Stage 1: Large Scale Pre-Training Data

4.1 阶段 1：大规模预训练数据

For pre-training, we use video-caption pairs from the licensed Shutterstock stock videos dataset¹, which contains 10M unique captions. Retaining one random clip per caption yields 10M video-caption pairs, equivalent to the WebVid-10M dataset [4] previously used in academic research.

对于预训练，我们使用了获得许可的 Shutterstock 股票视频数据集¹ 中的视频-字幕对，该数据集包含 1000 万个唯一的字幕。每个字幕保留一个随机片段，得到 1000 万个视频-字幕对，这相当于之前学术研究中使用的 WebVid-10M 数据集 [4]。



Original Caption

原始标题



Figure 4: images/3538ef4464218745e99d2fa20f0fbec825309a3c1e3986922d9ca7076384a15f.jpg



Mumbai/India 2.21.2016 video of Driving On Mumbai Roads taken by point of view camera
High Quality Claude-3-Sonnet Generated Caption

This video appears to be a first-person perspective drive through a quiet urban street in India during early morning hours. The camera moves along a wide, tree-lined road flanked by impressive multi-story buildings with ornate architecture. As the journey progresses, the sun rises, casting a warm glow on the buildings and empty street. Few vehicles and pedestrians are seen, creating a serene atmosphere as the city slowly awakens.

孟买/印度 2016 年 2 月 21 日视频，记录了在孟买道路上驾驶的视角

高质量 Claude-3-Sonnet 生成的字幕

这段视频似乎是一段清晨时分，以第一人称视角穿越印度安静城市街道的驾驶画面。摄像机沿着一条宽阔、两旁种有树木的道路行驶，道路两侧矗立着建筑精美、多层楼栋的建筑。随着旅程的推进，太阳升起，为建筑物和空旷的街道洒下温暖的光辉。可以看到少量的车辆和行人，营造出一种宁静的氛围，城市正缓缓苏醒。

Figure 4: Comparing original caption against Claude-3-Sonnet generated high-quality caption.

图 4：比较原始标题与 Claude-3-Sonnet 生成的高质量标题。

4.2 Stage 2: High Quality Intermediate Data

4.2 阶段 2：高质量中间数据

We use a re-captioned subset of the Shutterstock dataset, whose extensively labeled keywords we utilize for the balancing operation. We exclude keywords with < 30 occurrences. Starting with the most frequent remaining keywords, we add a maximum of 500 videos per keyword to the candidate pool. Further details are in Supplementary L. In order to generate this data within a limited budget, we generate captions for a diverse subset of

我们使用了一个重新生成标题的 Shutterstock 数据集的子集，我们利用其中广泛标注的关键词来进行平衡操作。我们排除了出现次数少于 30 次的关键词。从剩余的最频繁关键词开始，我们为每个关键词最多添加 500 个视频到候选池中。更多细节请参见补充材料 L。为了在有限的预算内生成这些数据，我们为一个多样化的子集生成了标题。

200,000 videos. We follow the insight from prior works (e.g., LLaVA [26], Meta-CLIP [48], Cambrian [40]) that under-sampling common concepts is important for large scale vision-language training to achieve better generalization on rare concepts.

200,000 个视频。我们遵循先前工作的见解（例如，LLaVA [26]、Meta-CLIP [48]、Cambrian [40]），认为在大规模视觉-语言训练中，对常见概念进行欠采样对于在罕见概念上实现更好的泛化能力非常重要。

We use the Claude-3-Sonnet model prompted with 20 frames from the video, each resized to 480 pixels height, along with the following text prompt: “Provide detailed narrative caption for the video, covering its overall theme, visuals, characters, actions, scene, and key moments comprehensively”. The original caption and generated high quality caption for a sample video are shown in Fig. 4.

我们使用了由视频中 20 帧图像提示的 Claude-3-Sonnet 模型，每帧图像的高度被调整为 480 像素，并结

合以下文本提示：“为视频提供详细的叙述式标题，涵盖其整体主题、视觉效果、角色、动作、场景和关键时刻”。一个示例视频的原始标题和生成的高质量标题如图 4 所示。

4.3 Stage 3: Multi-Task Fine-Tuning Data

4.3 第三阶段：多任务微调数据

Our fine-tuning stage is analogous to Instruction

我们的微调阶段类似于指令

FineTuning (IFT) in LLMs. We design our fine-tuning dataset to include multi-task supervision covering these tasks as well as

在大语言模型中的微调（IFT）。我们设计我们的微调数据集，以包含覆盖这些任务的多任务监督，以及

Dataset	#	Cap	QA	Loc
MSR-VTT	10K	□		□ □
ActivityNet	30K	□		□ □
DiDeMo	10K	□		□ □
Shutterstock	50K	□		□ □

数据集	#	Cap	QA	Loc
MSR-VTT	10K	□		□ □
ActivityNet	30K	□		□ □
DiDeMo	10K	□		□ □
Shutterstock	50K	□		□ □

Table 2: Multi-Task Dataset.

表 2: 多任务数据集。

video captioning. We select 100,000 samples from a small number of video-text datasets and obtain supervision from each as summarized in Table 2. All these datasets contribute to captioning, retrieval and matching supervision. MSR-VTT provides short video captions and question answering supervision. ActivityNet and DiDeMo consist of longer videos with distinct temporally localizable captions for chunks of the video. Hence, these datasets can provide localization supervision. ActivityNet has also previously been annotated with question answering data. We add a random subset of Shutterstock video-caption pairs (original captions) to the mix to maintain long tail diversity.

视频字幕生成。我们从少量的视频-文本数据集中选取了 100,000 个样本，并从每个数据集获得监督信息，如表 2 所示。所有这些数据集都为字幕生成、检索和匹配监督提供了贡献。MSR-VTT 提供了短视频字幕和问答监督。ActivityNet 和 DiDeMo 包含更长的视频，这些视频具有针对视频片段的不同时间可定位的字幕。因此，这些数据集可以提供定位监督。ActivityNet 之前还被标注了问答数据。我们添加了一个随机子集的 Shutterstock 视频-字幕对（原始字幕）以保持长尾多样性。

5 Results

5 结果

First we demonstrate the unified capabilities of our model on 8 benchmark datasets. Next up, we demonstrate the unique capabilities of our model by solving novel tasks that are hard for any single model prior to this work. Finally, we show ablations demonstrating the effectiveness of each component in our modeling and training approach.

首先，我们在 8 个基准数据集上展示了我们模型的统一能力。接下来，我们通过解决之前任何单一模型都难以处理的新任务，展示了我们模型的独特能力。最后，我们展示了消融实验，以证明我们建模和训练方法中每个组件的有效性。

5.1 Retrieval Tasks

5.1 检索任务

We evaluate our model on video retrieval, and moment retrieval/temporal localization tasks. These benchmarks together provide a holistic evaluation of video understanding capabilities. We use standard evaluation protocols following prior works [36, 37, 24].

我们在视频检索和片段检索/时间定位任务上评估我们的模型。这些基准测试共同提供了对视频理解能力的全面评估。我们采用标准的评估协议，遵循先前的工作 [36, 37, 24]。

Method	Text to Moment Retrieval			Text to Video Retrieval		
ActivityNet R@1, IoU=0.5	Charades R@1, IoU=0.5	QVHighlights mAP	MSR-VTT R@1	DiDeMo R@1	VATEX R@1	
Expert Models						
M-DETR NeurIPS 21	-	53.6 ^{FT}	35.7 ^{FT}	□	□	□
QD-DETR CVPR 23	33.2 ^{FT}	57.3 ^{FT}	38.9 ^{FT}	□	□	□
TR-DETR AAAI 24	-	57.6 ^{FT}	39.9 ^{FT}	□	□	□
SigLIP ICCV 23	□	□	□	51.7 ^{FT}	55.4 ^{FT}	40.8
VidLA CVPR 24	□	□	□	58.0 ^{FT}	61.1 ^{FT}	-
InternVideo2-□ CLIP-1B ECCV 24		□	□	50.0	47.7	63.2
Video LLMs						
Momentor- 7B ICML 24	23.0	27.5	7.6	□	□	□
VTimeLLM- 7B CVPR 24	27.8	27.5	-	□	□	□

TimeChat-7B CVPR 24	16.3	32.2	21.7	□	□	□
Qwen2.5-VL-3B arXiv	28.6	38.1	-	□	□	□
LLaVA-ST-7B CVPR 25	31.2	44.8	-	□	□	□
ViLL-E-2.5B (Ours)	39.4±8.2	51.5±6.7	39.0	62.5±4.5	61.4	63.5

方法	时刻检索			视频检索		
ActivityNet R@1, IoU=0.5	Charades R@1, IoU=0.5	QVHighlights mAP	MSR-VTT R@1	DiDeMo R@1	VATEX R@1	
专家模型						
M-DETR NeurIPS 21	-	53.6 ^{FT}	35.7 ^{FT}	□	□	□
QD-DETR CVPR 23	33.2 ^{FT}	57.3 ^{FT}	38.9 ^{FT}	□	□	□
TR-DETR AAAI 24	-	57.6 ^{FT}	39.9 ^{FT}	□	□	□
SigLIP ICCV 23	□	□	□	51.7 ^{FT}	55.4 ^{FT}	40.8
VidLA CVPR 24	□	□	□	58.0 ^{FT}	61.1 ^{FT}	-
InternVideo2-□ CLIP-1B ECCV 24	□	□	□	50.0	47.7	63.2
视频大语言模型						
Momentor-7B ICML 24	23.0	27.5	7.6	□	□	□
VTimeLLM-7B CVPR 24	27.8	27.5	-	□	□	□
TimeChat-7B CVPR 24	16.3	32.2	21.7	□	□	□
Qwen2.5-VL-3B arXiv	28.6	38.1	-	□	□	□
LLaVA-ST-7B CVPR 25	31.2	44.8	-	□	□	□
ViLL-E-2.5B (我们的)	39.4±8.2	51.5±6.7	39.0	62.5±4.5	61.4	63.5

Table 1: Results on retrieval tasks. In moment retrieval, we outperform specialized Video LLMs by over 8%, and approach the performance of SotA dataset-specific expert models. In video retrieval we beat finetuned SotA models. ^{FT} models fine-tuned on train splits. X- model is incapable of task

表 1: 检索任务的结果。在时刻检索任务中, 我们的表现优于专门的视频语言模型超过 8%, 并且接近领域专家模型的性能。在视频检索任务中, 我们超越了微调后的 SotA 模型。^{FT} 模型是在训练集上微调的。X-模型无法完成任务

Moment Retrieval: We test on ActivityNetCaptions, Charades-STA and QVHighlights. Contextualized text embeddings are extracted by passing {sparse video frames} <|txt_embed|> {text}. Video clips are extracted using a sliding window operation and clip embeddings are generated by passing {clip frames} <|vid_embed|>. Text and Video Clip embedding similarities are post-processed using standard methods to get the matching segment. Video Retrieval: We evaluate on MSR-VTT, DiDeMo and VATEX. Video embeddings are extracted by passing following input to the model: {video} <|vid_embed|>, while text embeddings are extracted by passing <|txt_embed|> {text}. Best pair is matched using nearest neighbor retrieval. In moment retrieval (Table 1), ViLL-E outperforms localization specialized video-LLMs by nearly 10%, and approaches or surpasses the per-dataset fine-tuned performance of non-LLM methods. In text-video retrieval, ViLL-E matches image (SigLIP) and video (VidLA) retrieval models.

时刻检索: 我们在 ActivityNetCaptions、Charades-STA 和 QVHighlights 上进行测试。上下文化的文本嵌入是通过将 {稀疏视频帧} <|txt_embed|> {文本} 传递给模型得到的。视频片段是通过滑动窗口操作提取的, 片段嵌入是通过将 {片段帧} <|vid_embed|> 传递给模型生成的。文本和视频片段嵌入的相似性使用标准方法进行后处理, 以获取匹配的片段。视频检索: 我们在 MSR-VTT、DiDeMo 和 VATEX 上进行评估。视频嵌入是通过将以下输入传递给模型得到的: {视频} <|vid_embed|>, 而文本嵌入是通过将 <|txt_embed|> {文本} 传递给模型得到的。最佳配对是通过最近邻检索进行匹配的。在时刻检索 (表 1) 中, ViLL-E 在定位专用视频-LLMs 上表现优异, 性能高出近 10%, 并且接近或超越了非 LLM 方法在每个数据集上的微调性能。在文本-视频检索中, ViLL-E 与图像 (SigLIP) 和视频 (VidLA) 检索模型表现相当。

5.2 QA Tasks

5.2 QA 任务

Among VideoLLMs specialized for temporal localization, our method achieves top performance in several categories (MSR, VCG, MVBench) indicated by green upward arrows showing improvements over previous methods. It slightly underperforms compared to LLaVA-ST on the MSVD dataset, indicated by a small red downward arrow. While SotA general VideoLLMs Qwen-2.5VL and LLaVA-Video outperform ViLL-E on MVBench & VideoMME, our approach remains competitive.

在专门针对时间定位的视频大模型中, 我们的方法在多个类别 (MSR、VCG、MVBench) 中表现最佳, 绿色向上的箭头表示相比之前的方法有所提升。与 LLaVA-ST 相比, 在 MSVD 数据集上我们的方法略有不足, 用一个小的红色向下的箭头表示。虽然目前最先进的通用视频大模型 Qwen-2.5VL 和 LLaVA-Video 在 MVBench 和 VideoMME 上优于 ViLL-E, 但我们的方法仍然具有竞争力。

5.3 Additional Tasks

5.3 其他任务

In this section we demonstrate ViLL-E’s ability to solve some unique tasks that are made possible by

在本节中, 我们展示了 ViLL-E 解决一些独特任务的能力, 这些任务得以实现得益于

Method	MSR	VCG	MSVD	MVBench	VidMME
General Video LLMs					
Qwen-2.5VL	-	-	-	67.0	61.5

LLaVA-Video	-	3.5	-	58.6	63.3
Temporally-Specialized Video LLMs					
ST-LLM	63.2	3.15	74.6	54.9	37.9
LLaVA-ST	59.0	3.3	75.9	64.2	
Momentor	55.6	3.0	68.9	-	-
VTimeLLM	50.2	2.9	-	-	-
TimeChat	45.0	2.3	-	38.5	34.7
ViLL-E	65.2	3.7	75.2	64.7	45.0

方法	MSR	VCG	MSVD	MVBench	VidMME
通用视频大语言模型					
Qwen-2.5VL	-	-	-	67.0	61.5
LLaVA-Video	-	3.5	-	58.6	63.3
时间专用视频大语言模型					
ST-LLM	63.2	3.15	74.6	54.9	37.9
LLaVA-ST	59.0	3.3	75.9	64.2	
Momentor	55.6	3.0	68.9	-	-
VTimeLLM	50.2	2.9	-	-	-
TimeChat	45.0	2.3	-	38.5	34.7
ViLL-E	65.2	3.7	75.2	64.7	45.0

Table 3: ViLL-E is competitive with SotA VideoLLMs on QA tasks in both simple VideoQA (MSR-VTT QA, VideoChatGPT Benchmark and MSVD-QA) and Comprehensive Benchmarks (MVBench and VideoMME).

having unified generative and embedding capabilities in one VideoLLM.

表 3: ViLL-E 在简单视频问答 (MSR-VTT QA、VideoChatGPT 测试集和 MSVD-QA) 和综合基准测试 (MVBench 和 VideoMME) 中, 在问答任务上与 SotA 视频语言模型具有竞争力。在一个 VideoLLM 中具备统一的生成能力和嵌入能力。

Two Stage Retrieval using Re-Ranking: This task leverages the VideoLLM’s reasoning capability to augment retrieval performance. In the first stage, Top-K candidate videos matching a caption are retrieved using generated embeddings. During second stage matching, each candidate videocaption pair is input to the VideoLLM and the matching loss is used to re-rank the Top-K candidate videos. The results in Table 4 (K = 25) show that our two stage matching process results in an improvement of about 2% in R@1 accuracy.

两阶段检索使用重排序: 该任务利用 VideoLLM 的推理能力来提升检索性能。在第一阶段, 通过生成的嵌入向量检索与描述匹配的前 K 个候选视频。在第二阶段匹配中, 每个候选视频-描述对输入到 VideoLLM 中, 并利用匹配损失对前 K 个候选视频进行重排序。表 4 (K = 25) 的结果显示, 我们的两阶段匹配过程使 R@1 准确率提高了约 2%。

MSR-VTT	T $\rightarrow$ V		V $\rightarrow$ T	
	R@1	R@5	R@1	R@5
VidLA	58.0	81.1	56.1	80.5

Ours (1 Stage)	62.5	78.1	55.3	74.8
Ours (2 Stage)	62.8	80.1	57.3	83.5

MSR-VTT	T $\rightarrow$ V		V $\rightarrow$ T	
方法	R@1	R@5	R@1	R@5
VidLA	58.0	81.1	56.1	80.5
我们的 (1 阶段)	62.5	78.1	55.3	74.8
我们的 (2 阶段)	62.8	80.1	57.3	83.5

Table 4: Two-stage retrieval + matching.

表 4: 两阶段检索 + 匹配。

Composed Video Retrieval [42] is the task of retrieving a video matching a given query video along with a modifier text. The retrieved video

组合视频检索 [42] 是一项任务，其目标是根据给定的查询视频和一个修饰文本，检索出与之匹配的视频。检索到的视频

must closely match a hypothetical video containing the contents of the original video further modified by the instructions in the change text. This is a relatively new task in the field of video retrieval and forms a challenging zero-shot benchmark for our model. To solve this, we extract multi-modal query embeddings by passing {source video frames} <|txt_embed|> {change text}, and retrieve videos by matching with target video embeddings extracted as we do for retrieval. The results in Table 5 demonstrate that in the zero-shot setting our method surpasses recent baselines by more than 5%.

必须紧密匹配一个假设的视频，其中包含原始视频的内容，并进一步经过变更文本中指令的修改。这是视频检索领域相对较新的任务，并为我们的模型形成一个具有挑战性的零样本基准。为了解决这个问题，我们通过传递 {source video frames} <|txt_embed|> {change text} 来提取多模态查询嵌入，并通过与我们用于检索的目标视频嵌入进行匹配来检索视频。表 5 中的结果表明，在零样本设置下，我们的方法优于最近的基线模型超过 5%。

Method	R@1	R@5	R@10
COVR-BLIP AAAI 24	45.46	70.46	79.54
Thawakar et al. CVPR 24	47.52	72.18	82.37
Ours	53.13	74.80	85.95

方法	R@1	R@5	R@10
COVR-BLIP AAAI 24	45.46	70.46	79.54
Thawakar 等人 CVPR 24	47.52	72.18	82.37
我们的	53.13	74.80	85.95

Table 5: Zero-Shot Composed Video Retrieval.

表 5: 零样本组合视频检索。

Detailed Video Caption Retrieval: Dual encoder video-text retrieval models, like CLIP, are limited by short context windows (e.g., 77 tokens). Our LLM approach supports much longer texts:

详细视频字幕检索：像 CLIP 这样的双编码器视频-文本检索模型受到短上下文窗口（例如 77 个标记）的限制。我们的 LLM 方法支持更长的文本：

Method	Short CaptionAvg. = 27 words				Long CaptionAvg. = 500 words			
	T $\rightarrow$ V		V $\rightarrow$ T		T $\rightarrow$ V		V $\rightarrow$ T	
	R@1	R@5	R@1	R@5	R@1	R@5	R@1	R@5
SigLIP ICCV23	62.5	83.2	65.4	85.1	51.7	72.7	60.4	81.1
VidLA CVPR24	61.2	85.7	65.9	85.0	45.3	65.2	55.9	72.7
LongCLIP ECCV24	63.0	85.4	61.5	83.6	73.5	91.7	74.5	92.4
ViLL-E	64.3	85.7	65.5	85.4	75.7	92.4	75.1	93.1

方法	简短字幕平均长度 = 27 字				长标题平均长度 = 500 字			
	T $\rightarrow$ V		V $\rightarrow$ T		T $\rightarrow$ V		V $\rightarrow$ T	
	R@1	R@5	R@1	R@5	R@1	R@5	R@1	R@5
SigLIP ICCV23	62.5	83.2	65.4	85.1	51.7	72.7	60.4	81.1
VidLA CVPR24	61.2	85.7	65.9	85.0	45.3	65.2	55.9	72.7
LongCLIP ECCV24	63.0	85.4	61.5	83.6	73.5	91.7	74.5	92.4
ViLL-E	64.3	85.7	65.5	85.4	75.7	92.4	75.1	93.1

Table 6: Zero-Shot Video Detailed Caption Retrieval. thousands of tokens, enabling effective zero-shot retrieval using detailed captions. Existing benchmarks (e.g., YouCook2) are either too domain-specific or overlap with our training data. We use the AuroraCap-VideoDetailCaption dataset [2], which includes both long and short captions for 1,027 videos. As shown in Table 6, our model benefits from longer captions, improving R@1 by 11.4% (text-to-video) and 9.6% (video-to-text). In contrast, models like SigLIP [53] and VidLA [37] are constrained by short text encoder contexts (64 tokens). LongCLIP [54], with a 248-token limit, improves on long captions but lags our approach.

表 6：零样本视频详细描述检索。数千个标记，使得能够使用详细描述进行有效的零样本检索。现有的基准测试（例如 YouCook2）要么过于特定于某个领域，要么与我们的训练数据重叠。我们使用 AuroraCap-VideoDetailCaption 数据集 [2]，其中包含 1,027 个视频的长和短描述。如表 6 所示，我们的模型受益于更长的描述，使文本到视频的 R@1 提高了 11.4%，视频到文本的 R@1 提高了 9.6%。相比之下，像 SigLIP[53] 和 VidLA[37] 这样的模型受到短文本编码器上下文（64 个标记）的限制。LongCLIP[54] 虽然在长描述方面有所改进，但其限制在 248 个标记，仍落后于我们的方法。

5.4 Ablations

5.4 消融实验

We ablate the design choices for key components of our approach. For computational efficiency, some experiments are performed without pre-training.

我们对方法中关键组件的设计选择进行了消融实验。为了计算效率，一些实验未进行预训练。

Supervision Types: (Table 7a) We find that combining generative and contrastive objectives during fine-tuning complements each other, resulting in improvement on both generative and embeddings tasks. Retrieval and localization performances drop

监督类型：（表 7a）我们发现，在微调过程中结合生成式和对比目标可以相辅相成，从而在生成任务和嵌入任务上都取得改进。检索和定位性能下降

G	C	M	MSR	VCG	MSR	DDM	VTX	ANet	QVH	Ch.
□	□	□	65.1	3.7	62.8	61.5	63.7	39.4	38.9	51.7
□	□	×	63.9	3.7	60.3	60.2	62.2	39.1	38.5	51.5
□	×	×	61.3	3.0	25.1	23.5	11.9	28.7	23.8	42.1
×	□	×	45.5	2.1	54.7	53.1	55.2	29.3	30.3	48.8

G	C	M	MSR	VCG	MSR	DDM	VTX	ANet	QVH	章
□	□	□	65.1	3.7	62.8	61.5	63.7	39.4	38.9	51.7
□	□	×	63.9	3.7	60.3	60.2	62.2	39.1	38.5	51.5
□	×	×	61.3	3.0	25.1	23.5	11.9	28.7	23.8	42.1
×	□	×	45.5	2.1	54.7	53.1	55.2	29.3	30.3	48.8

(a) Supervision Type (During Finetuning)

(a) 监督类型（微调期间）

Head	MSR	VCG	MSR	DDM	VTX	ANet	QVH	Ch.
Ours	55.9	3.1	49.3	45.5	45.3	32.3	32.6	49.3
×	40.8	2.8	30.7	29.1	32.1	19.8	20.4	30.7
Linear	42.9	2.8	32.4	29.3	33.2	23.6	25.4	42.7
MLP	51.7	3.0	43.5	39.8	39.5	30.2	29.7	47.1
Self-Att	52.5	3.2	43.8	40.9	40.9		28.2	45.7
Q-Former	52.1	3.0	47.5	43.2	42.8		27.9	44.2
K-Former	52.7	3.1	49.0	44.1	44.7		32.8	47.4

头	MSR	VCG	MSR	DDM	VTX	ANet	QVH	章
我们的	55.9	3.1	49.3	45.5	45.3	32.3	32.6	49.3
×	40.8	2.8	30.7	29.1	32.1	19.8	20.4	30.7
线性	42.9	2.8	32.4	29.3	33.2	23.6	25.4	42.7
多层感知机（Multilayer Perceptron）	51.7	3.0	43.5	39.8	39.5	30.2	29.7	47.1
自注意力（Self-Attention）	52.5	3.2	43.8	40.9	40.9		28.2	45.7
Q-Former	52.1	3.0	47.5	43.2	42.8		27.9	44.2
K-Former	52.7	3.1	49.0	44.1	44.7		32.8	47.4

(b) Embedding Head Design

(b) 嵌入头设计

Pre HQ		MSR	VCG	MSR	DDM	VTX	ANet	QVH	Ch.
□	□	65.1	3.7	62.8	61.5	63.7	39.4	38.9	51.7
□	×	63.2	3.3	58.6	59.6	60.5	36.8	38.8	51.9
×	×	55.9	3.1	49.3	45.5	45.3	32.3	32.6	49.3

Pre HQ		MSR	VCG	MSR	DDM	VTX	ANet	QVH	章
□	□	65.1	3.7	62.8	61.5	63.7	39.4	38.9	51.7
□	×	63.2	3.3	58.6	59.6	60.5	36.8	38.8	51.9
×	×	55.9	3.1	49.3	45.5	45.3	32.3	32.6	49.3

(c) Impact of Pre-Training
Table 7: Ablation Experiments.

(c) 预训练的影响
表 7: 消融实验。

Benchmarks (left to right). QA: MSR-VTT-QA, VideoChat-GPT. Retrieval: MSR-VTT, DiDeMo, VATEX. Localization: ActivityNet, QVHighlights, Charades-STA.

基准测试（从左到右）。问答：MSR-VTT-QA, VideoChat-GPT。检索：MSR-VTT, DiDeMo, VATEX。定位：ActivityNet, QVHighlights, Charades-STA。

Legend: (a) G = Generative, C = Contrastive, M = Matching. (c) Pre. = 10M scale; HQ = High-Quality 200k.

图例：(a) G = 生成式, C = 对比式, M = 匹配式。(c) Pre. = 10M 规模; HQ = 高质量 200k。

moderately on removing matching loss, and significantly on removing contrastive training, indicating the importance of learning good embeddings.

在去除匹配损失方面有适度的提升，在去除对比训练方面有显著的提升，这表明学习良好的嵌入向量的重要性。

Embedding Head (Table 7b) We find that our attentive pooling head outperforms other potential approaches discussed previously in Section 3.2.

嵌入头（表 7b）我们发现，我们的注意力池化头在之前第 3.2 节讨论的其他潜在方法中表现更优。

Pre-Training (Table 7c) The large scale pre-training stage is very important for retrieval tasks, whereas the second pre-training stage with longer higher quality captions helps particularly in longer video datasets and generative tasks.

预训练（表 7c）大规模的预训练阶段对于检索任务非常重要，而使用更长、质量更高的描述文本进行的第二阶段预训练，尤其在处理较长的视频数据集和生成性任务时有所帮助。

Additional experiments and analysis of our adaptive embedding mechanism: including inference latency, per-duration-bin retrieval accuracy, and distribution of generated token counts, are provided in Appendix A (Tables 10 and 11).

附加实验和分析我们的自适应嵌入机制：包括推理延迟、每持续时间桶检索准确率以及生成的标记计数分布，详见附录 A（表 10 和表 11）。

6 Conclusion

6 结论

We introduced ViLL-E, a unified VideoLLM that combines text generation and embedding-based retrieval within one model. Through a multi-stage generative-contrastive training strategy and an adaptive embedding head, ViLL-E achieves strong results across retrieval, localization, and QA, while enabling new zero-shot tasks like composed and long-caption retrieval. This work highlights the potential of merging generative and discriminative learning for unified multimodal understanding.

我们引入了 ViLL-E，这是一个统一的 VideoLLM，它在一个模型中结合了文本生成和基于嵌入的检索。通过多阶段生成-对比训练策略和自适应嵌入头，ViLL-E 在检索、定位和问答任务中取得了优异的性能，同时支持诸如组合型和长描述检索等新的零样本任务。这项工作突显了将生成式学习和判别式学习相结合在统一多模态理解中的潜力。

7 Limitations

7 局限性

Our model inherits some limitations of the base Paligemma LLM, e.g. it lacks general multi-turn conversation abilities. Secondly, because our training dataset is primarily in English, we expect that our model would lose some of Paligemma’s multilingual capabilities. Our work serves as a proof of concept for the unification of generative and embedding capabilities in VideoLLMs, and any practical model would need to address the issues of multi-turn conversations and multilinguality.

我们的模型继承了基础 Paligemma 大语言模型的一些限制，例如它缺乏一般的多轮对话能力。其次，由于我们的训练数据集主要以英语为主，我们预计我们的模型会失去 Paligemma 的多语言能力。我们的工作为 VideoLLMs 中统一生成能力和嵌入能力提供了一个概念验证，任何实际的模型都需要解决多轮对话和多语言性的问题。

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